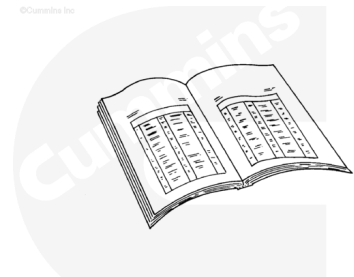


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the coolant and remove the radiator. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>).
- Remove the fan drive belt, if applicable. Refer to Procedure 008-002 in Section 8

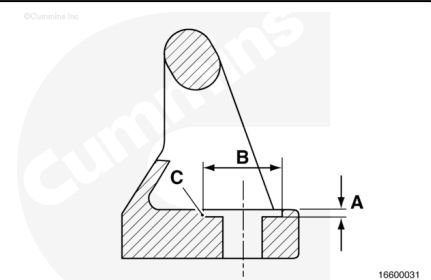
(/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html).

- Remove the vibration damper. Refer to Procedure 001-052 in Section 1 (/qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html).
- Remove the crankshaft adapter. Refer to Procedure 001-017 in Section 1 (/qs3/pubsys2/xml/en/procedures/28/28-001-017-tr.html).

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

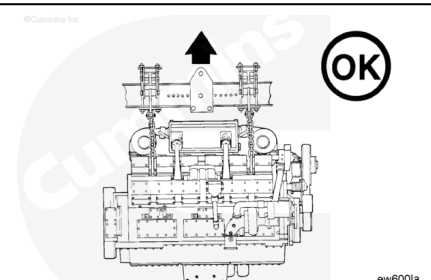


⚠ WARNING ⚠

Before lifting the engine, check the four engine lifting brackets. Be sure bracket contains a spot face under the hardened steel washer. If the bracket is not spot faced, remove the capscrew.

Check for corrosion. If the capscrew has corrosion, replace the capscrew before lifting the engine.

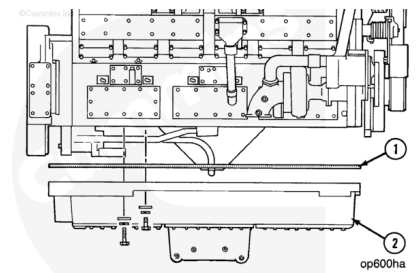
Support the engine in strict accordance with correct safety considerations and precautions. Refer to Procedure 000-001 in Section 0 (/qs3/pubsys2/xml/en/procedures/28/28-000-001.html).



If the front of the engine **must** be supported from underneath while work is performed on the front end, drain the oil and remove the oil pan adapter assembly first. Refer to Procedure

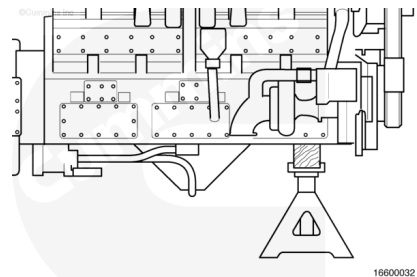
007-027 in Section 7

(/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html).

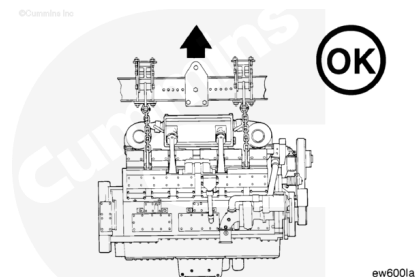


The engine can then be supported by a heavy wooden timber placed under the block pan rails.

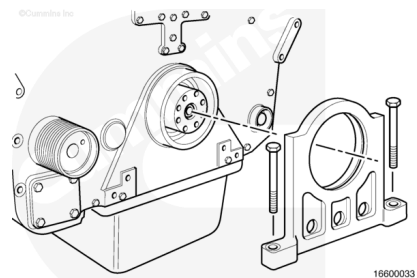
Use suitable means to take the weight off the front support.



K38 and K50 engines have four lifting brackets that are bolted to the cylinder block. Lifting brackets are designed to lift the engine only.



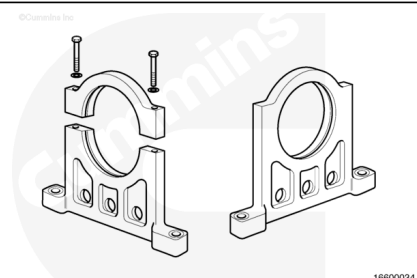
Remove the capscrews that mount the front support to the chassis or frame. Remove the front support from the gear housing cover trunnion.



Inspect for Reuse

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Damaged engine mounts and brackets can cause the engine to move out of alignment, damage the driveline components in the equipment, and result in vibration complaints.

Two different front support assemblies have been supplied on K38 and K50 engines. Earlier engines have a two-piece support. Later engines use a one-piece support.

Inspect the front engine mount for cracks or other damage.

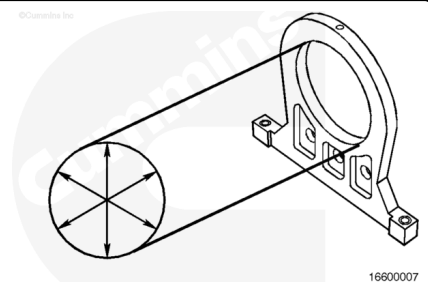
Inspect all mounting brackets for cracks or damaged bolt holes.

Replace the bracket if cracked or otherwise damaged.

Measure the front support bore.

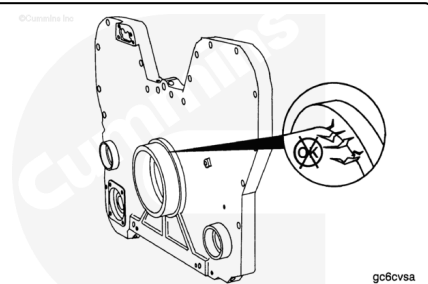
Torque Value: 300 n•m [221 ft-lb]

Front Support Bore Diameter		
mm		in
210.03	MIN	8.269
210.08	MAX	8.271



If the front support bore is **not** within specifications, the front engine support support bracket **must** be replaced.

Check the trunnion sleeve for damage. The sleeve **must** be replaced if it is worn or cracked.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this

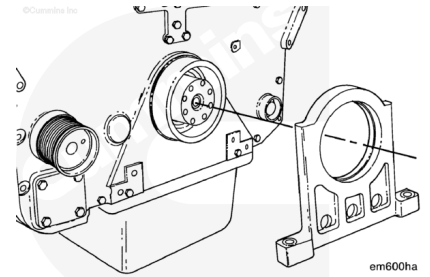
component or assembly.

Use high performance, industrial-type grease (NLGI Grade Number 2) fortified with extreme pressure (EP) additives, such as BP Energrease™ FPG or Castrol Spheerol™ EPL 2 grease, or its equivalent.

If the above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

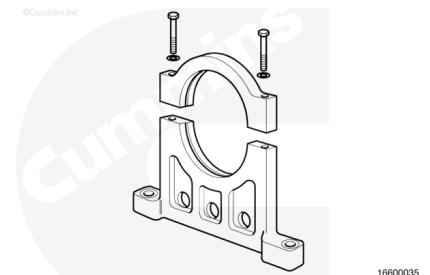
Lubricate the bore of the mount and trunnion sleeve on the front cover gear.

Install the front engine support by slipping it over the trunnion bushing.

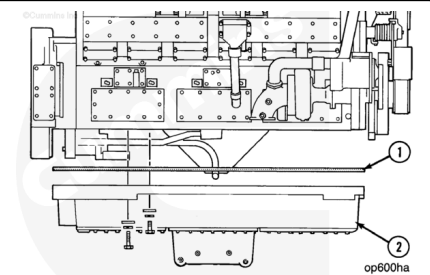


If the two-piece front support is used, tighten the capscrews to the specified torque.

Torque Value: 300 n•m [221 ft-lb]



Raise the engine to take the weight off the supports used under the engine block. Remove the supports and install the oil pan. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html>)

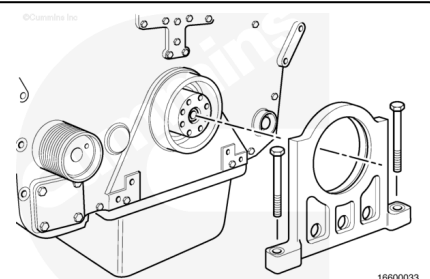


Lower the engine until the front engine support is in position.

Remove the engine lifting fixture.

See equipment manufacturer service information for torque specifications.

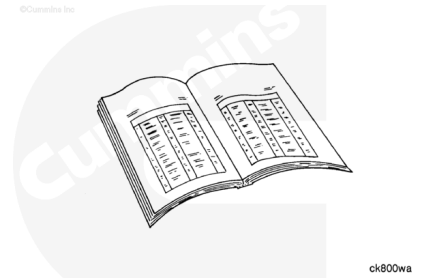
Remove engine lifting fixture.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

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⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the crankshaft adapter. Refer to Procedure 001-017 in Section 1 (</qs3/pubsys2/xml/en/procedures/28/28-001-017-tr.html>).
- Install the vibration dampers. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html>).
- Install fan drive belt, if applicable. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>).
- Replace items removed and fill with coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>).
- Connect the batteries. See equipment manufacturer service information.

